THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS CHAPTER TEST SET A

Chapter 2: **Power Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. $2^5 \times 2^3$ equals: 1

(a) 2^{15} **(b)** 4^8 **(c)** 2^8 **(d)** 2^2

2. $10^6 \times 10^4$ equals: 1

(a) 10^{10} **(b)** 10^2 **(c)** 10^{24} **(d)** 100^{10}

3. $7^{12} \div 7^5$ equals: 1

(a) 7^7 **(b)** 7^2 **(c)** 7^{17} **(d)** 7^{60}

4. $(3^4)^2$ equals: 1

(a) 3^{16} **(b)** 3^6 **(c)** 12^2 **(d)** 3^8

5. The value of 5^{-2} is: 1

(a) 25 **(b)** $\frac{1}{25}$ **(c)** -25 **(d)** $\frac{1}{10}$

6. The value of $(-7)^0$ is: 1

(a) 0 **(b)** −7 **(c)** 1 **(d)** 7

7. The number 72000 in standard (scientific) form is: 1

(a) 0.72×10^4 **(b)** 7.2×10^3 **(c)** 72×10^4 **(d)** 7.2×10^4

8. The number 150000000 in standard (scientific) form is: 1

(a) 1.5×10^8 **(b)** 1.5×10^9 **(c)** 1.5×10^7 **(d)** 15×10^7

9. One billion is: 1

(a) 10^7 **(b)** 10^6 **(c)** 10^9 **(d)** 10^{12}

10. The value of $\left(\frac{1}{3}\right)^{-2}$ is: 1

(a) 6 **(b)** $\frac{1}{9}$ **(c)** $-\frac{1}{9}$ **(d)** 9

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Simplify $2^5 \times 2^{-3}$. 2

12. Write 5^{-3} as a fraction. 2

13. Express 0.00045 in scientific notation. 2

14. Write 6.2×10^5 in usual form. 2

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SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Simplify: $\dfrac{2^4 \times 2^3}{2^5}$. **3**

16. Simplify: $\dfrac{3^5 \times 3^2}{3^6}$. **3**

17. Simplify: $\dfrac{5^3 \times 5^4}{5^2}$. **3**

OR

Simplify: $\dfrac{7^4 \times 7^2}{7^3}$. **3**

18. Simplify: $\dfrac{10^8 \times 10^5}{10^4}$. **3**

OR

Simplify: $\dfrac{2^6 \times 2^5}{2^2}$. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) State the laws of exponents for $a^m \times a^n$, $a^m \div a^n$ and $(a^m)^n$. **5**

(ii) Simplify $\dfrac{2^7 \times 5^4}{10^3}$.

OR

(i) Write 4.7×10^{-4} in decimal form. **5**

(ii) Write 0.00000092 in standard form.

(iii) Add $3 \times 10^4 + 5 \times 10^3$ and express in standard form.

20. (i) Find the value of $\left(\frac{2}{3}\right)^{-3} \times \left(\frac{2}{3}\right)^2$. **5**

(ii) Simplify $\dfrac{(25)^3 \times 5^2}{5^{10}}$.

OR

Case study: A laboratory sample has 2×10^9 bacteria. The population triples every hour. **5**

(i) How many bacteria are there after 3 hours?

(ii) Express the answer in standard form.

(iii) After how many hours will it first exceed 10^{11}?

END OF QUESTION PAPER — Total: 40 Marks

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